Arabic Toxicity Detection & Evaluation Across LLMs: Task 2

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I. Dataset

This study utilizes the **MAHED 2025 shared task** dataset [1]–[3], a recent benchmark for Arabic harmful and prosocial content analysis. The dataset is designed to support multiple related tasks, including text-based classification, multitask learning, and multimodal content understanding, enabling comprehensive evaluation of language models across different dimensions of toxicity and affective content.

In this work, we focus on the text-based components of the dataset, specifically Task 1 and Task 2. Task 1 formulates the problem as a three-class classification task (*hate*, *hope*, *not_applicable*), while Task 2 extends this setting to a multitask framework that jointly predicts emotion, offensiveness, and hate labels. The multimodal task (Task 3), which involves image-text memes, is excluded as this study focuses on text-only modeling.

The dataset is selected for several reasons. First, it captures both harmful (hate) and constructive (hope) language, allowing models to distinguish between negative, positive, and neutral content. Second, unlike other similar datasets such as [4], [5] which are either orginally curated for binary classification or sentiment analysis, MAHED 2025 goes beyond this by introducing structured and multi-dimensional annotations, particularly in Task 2. Third, it is constructed from real-world social media data, which reflects the linguistic variability of Arabic, including dialectal usage and informal writing. Finally, the dataset is annotated by native speakers with high inter-annotator agreement, which improves the reliability of the evaluation.

A. Data Source and Collection

The MAHED dataset is constructed from publicly available Arabic content collected from social media platforms, featuring a variety of user-generated text such as informal writing, dialectal variations, and noisy linguistic patterns commonly observed in online communication. In addition, The dataset captures diverse expressions of language, including code-switching, non-standard spelling, and varying levels of formality.

MAHED was constructed by combining several existing Arabic datasets into a single unified benchmark. This allows different tasks, such as classification and multitask prediction, to be defined over a shared data format. The use of multiple sources increases variation in writing style and dialect, while annotations are aligned under a consistent labeling scheme.

B. Annotation Process

The dataset was annotated by native Arabic speakers. The paper reports Cohen's Kappa above 0.85, which indicates high agreement between annotators.

For Task 1, each text is assigned one label: hate, hope, or not_applicable. For Task 2, each text is annotated with three labels: one emotion label, one offensiveness label, and one hate label for offensive content. In this task, hate is treated as a subset of offensive content.

C. Label Definitions

MAHED includes two text-based tasks with different label structures.

1) Task 1 Labels: Task 1 is a three-class classification task. Each text is assigned one of the following labels:

Hate: text that contains offensive, harmful, or discriminatory language directed toward a person or group based on protected characteristics such as religion, nationality, or ethnicity.

Hope: text that expresses positive and constructive content, such as encouragement, gratitude, optimism, or future-oriented support.

Not_applicable: text that does not contain explicit hate or hope. This class includes all remaining cases outside the first two categories.

2) Task 2 Labels: Task 2 uses three related annotations for each text.

Emotion: each text is assigned one label from 12 emotion categories: *neutral, anger, anticipation, disgust, fear, joy, love, optimism, pessimism, sadness, surprise,* and *trust*.

Offensive: this label indicates whether the text contains offensive language. It is assigned as either *yes* or *no*.

Hate: this label is applied within offensive content. If the text targets an identity group such as religion, nationality, ethnicity, or gender, it is labeled as *hate*.

Table I: Example annotations from MAHED Task 1.

Text	Label
كل المهاجرين لصوص ويجب طردهم فوراً	**hate**
معاً يمكننا بناء مستقبل أفضل لأطفالنا	**hope**
اليوم هو يوم مشمس وجميل	**not_applicable**

Table II: Example annotations from MAHED Task 2.

Text	Emotion	Offensive	Hate
كل المهاجرين لصوص ويجب طردهم	anger	yes	**hate**
يا حمار ليش نسيت المفتاح؟	anger	yes	**not_hate**
أشعر بالحزن لأني خسرت وظيفتي	sadness	**no**	—

Otherwise, it is labeled as *not_hate*. In other words, all hate speech is offensive, but not all offensive content is hate speech.

Tables I and II show representative examples from the two text-based tasks. These examples illustrate the difference between hate and hope in Task 1, and the relation between emotion, offensiveness, and hate in Task 2.

D. Descriptive Analysis

We further examine the two text-based tasks using basic descriptive statistics and linguistic characteristics. For Task 1, the official split contains 6,890 training samples, 1,476 validation samples, and 1,477 test samples. The label distribution is imbalanced, with *not_applicable* forming the largest class in all three splits, followed by *hope*, while *hate* is the smallest class. This pattern is shown in Figure 1. Such imbalance makes the task harder, since models must distinguish minority classes from a dominant background class.

Figure 3 shows the text length distribution for Task 1. Most samples are short, which is expected for social media text. The distribution is right-skewed, with many samples concentrated at lower word counts and a smaller number of longer texts. Table IV provides additional linguistic statistics. The average length is 22.38 words per sample. The dataset also contains common social media markers such as mentions, emojis, hashtags, and URLs, although their frequencies are lower than in Task 2. Informal writing patterns are also present, including repeated punctuation and character elongation. In addition, 37.36% of the samples contain Latin-script text, which suggests the presence of mixed-script writing or code-switching.

Task 2 is more structurally complex, since each sample is annotated with three related outputs: emotion, offensiveness, and hate. The official split contains 5,960 training samples, 1,277 validation samples, and 1,278 test samples. Figure 2 shows the distribution of emotion labels across

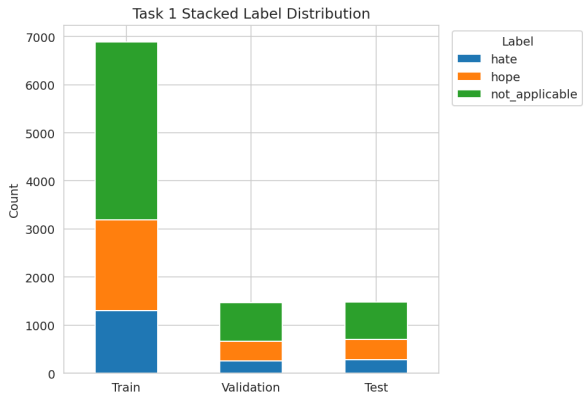

Figure 1: Label distribution across the train, validation, and test splits of MAHED Task 1. The figure shows that *not_applicable* is the largest class in all splits, while *hate* is the smallest.

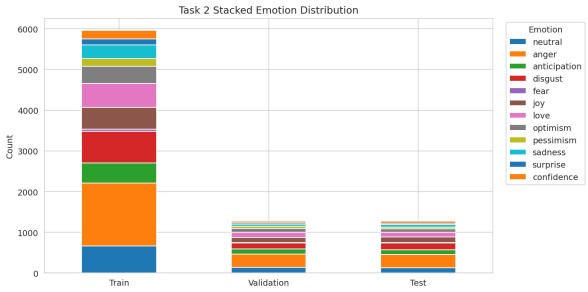

Figure 2: Emotion label distribution across the train, validation, and test splits of MAHED Task 2. The figure shows clear imbalance across emotion categories.

the three splits. The distribution is clearly imbalanced, with some emotions appearing much more frequently than others. Figure 4 shows the overall text length distribution. Similar to Task 1, most texts are short and the distribution is right-skewed.

The linguistic statistics of Task 2 are summarized in Table III. The average length is 21.15 words per sample, which is close to Task 1. However, Task 2 contains higher frequencies of social media markers and noisy writing patterns. Mentions appear in 50.84% of the samples, URLs in 29.67%, and emojis in 32.55%. Repeated punctuation and elongation are also common. A notable characteristic is the high proportion of Latin-script usage, which reaches 84.69%. This indicates that Task 2 includes a large amount of mixed-script content and potentially more code-switching than Task 1.

This descriptive statistical experiment reflect the nature of arabic social media text: short length, class imbalance, informal writing, and noisy platform-specific markers. These characteristics increase the difficulty of the benchmark and should be considered when interpreting model performance.

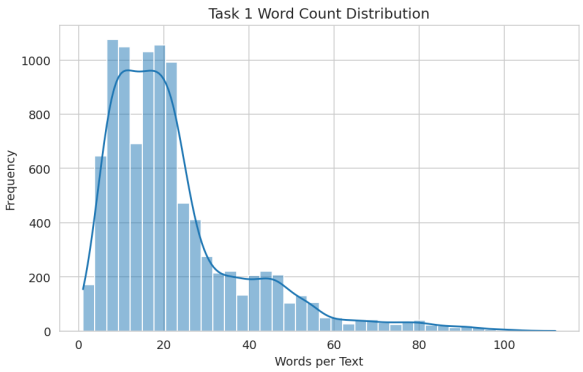

Figure 3: Distribution of text length in MAHED Task 1 measured by the number of words per sample. Most texts are short, with a right-skewed distribution and a small number of longer instances.

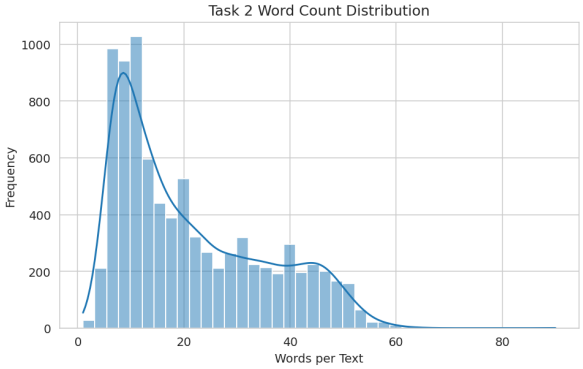

Figure 4: Label distribution across the train, validation, and test splits of MAHED Task 2.

E. Inter-Annotator Agreement

The MAHED paper reports a Cohen's Kappa score above 0.85 for the original annotation process, which indicates high agreement among the dataset annotators.

As a supplementary check, we randomly selected 20 samples from Task 1 and removed their gold labels. Table V lists the selected samples and their labels. Three annotators then labeled the samples independently using the Task 1 definitions. Table VI summarizes the agreement statistics, while Figure 5 shows the confusion matrices of each annotator against the dataset labels.

The results show noticeable variation across annotators. The overall Krippendorff's α is 0.326, and the pairwise Cohen's κ values range from 0.260 to 0.356. This indicates low to moderate agreement in the manual exercise. The disagreements were concentrated in a small number of recurring cases rather than being random.

The most common source of disagreement was *implicit hate*, where hostility toward a person or group was expressed without explicit insults, causing some samples to be labeled as *not_applicable* instead of *hate*. A second source of disagreement was the distinction between *political criticism* and *hate*, since strongly negative or ideological statements were sometimes interpreted as hate even when no direct target was present under the dataset guidelines. A third difficulty was the boundary between *hope* and *not_applicable*, especially in poetic, emotional, or religious text that expressed positive affect without direct motivational wording.

Because this experiment is based on only 20 samples, it should be viewed as a small sanity check rather than a replacement for the official dataset agreement score. Still, it helps show that the main difficulty in MAHED lies in the contextual interpretation.

Table III: Linguistic characteristics of the MAHED Task 2 dataset.

Metric	**Value**
Length	
Average word count	21.15
Social media markers	
Texts with emoji (%)	32.55
Texts with hashtag (%)	26.44
Texts with mention (%)	50.84
Texts with URL (%)	29.67
Informal writing patterns	
Texts with repeated punctuation (%)	17.51
Texts with elongation (%)	22.80
Texts with Latin/code-switching (%)	84.69

Table IV: Linguistic characteristics of the MAHED Task 1 dataset.

Metric	**Value**
Length	
Average word count	22.38
Social media markers	
Texts with emoji (%)	14.02
Texts with hashtag (%)	4.91
Texts with mention (%)	24.92
Texts with URL (%)	8.49
Informal writing patterns	
Texts with repeated punctuation (%)	15.24
Texts with elongation (%)	13.92
Texts with Latin/code-switching (%)	37.36

References

[1] W. Zaghouani, M. R. Biswas, M. Bessghaier, S. Ibrahim, G. Mikros, A. Hasnat, and F. Alam, "Mahed shared task: Multimodal detection of hope and hate emotions in arabic content," in *Proceedings of the third arabic natural language processing conference: shared tasks*, 2025, pp. 560–574.

[2] W. Zaghouani and M. R. Biswas, "An annotated corpus of arabic tweets for hate speech analysis," in *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing-Natural Language Processing in the Generative AI Era*, 2025, pp. 1413–1419.

[3] ——, "Emohopespeech: An annotated dataset of emotions and hope speech in english and arabic," in *Proceedings of the 15th*

Table V: Samples used in the supplementary manual agreement check for MAHED Task 1, along with their gold labels.

ID	Text	Label
8463	رحمه الله عليه سبحان الله عيناه تحكي قصه عطاء تنبع عيناه عطف وحنان [emojies] sd7n 1	hope
4038	2023/2/25 تاريخ مميز نهايه حقبة صاحب السوابق المشبوة مرتضي احمد منصور [URL] RT [USER]	hate
5824	انا حارجع العب چودو تاني و حاتمرن جامد و اروح الاولمبياد الاعب الاسرائيلي و اهزمه و لما يجي يسلم عليا حاسلم عليه و اعطس في وشه اغرقه تفافه	hate
5411	سكاي نيوز تزور فلما عن المرتزقه الروس في سوريا. للعلم، سكاي نيوز مملوكه للصهيوني روبرت مردوخ.	not_applicable
1284	يا صاح لست من الغرام بصاح مادامت الأرواح في الأشباح أتلوم بالحدق المراض وطالما تركت صحاح الناس غير صحاح روحي الفداء لمن أبيت وشعره ليلي وضوء جبينه مصباحي في خده وردي وفي أصداغه آسي وفي وجناته تفاحي وبريقه المعسول لا أبريقه راح المحب ونشوة المرتاح فوحقّ ما في ثغره من لؤلؤ رطب ومن مشمولة وأقاح ووحقّ ما في قده ولحاظه من ذابل لدن وبيض صفاح	hope
9204	جدهم الخميني كلهم مع إصحاب تعز واب [USER]	hate
3643	هنجيب ميداليه ف الاولمبياد يا عماد [emojies]!	hope
6046	انكشفت امريكا منذ عقود تتلاعب بالعالم وتختلق الحروب وتظهر المبررات وتدعي بخدمه البشرية والعالم يصدقها وظهرت حقيقتها لمن كان يصدقها عدوة للانسانية وسياسه الحرية وحقوق الانسان كانت تستغلها لتسيطر على العالم وتتلاعب بالشعارات لأهدافها والان هذه امريكا الشيطان على حقيقتها بدون زيف [USER]	not_applicable
8563	اللهم اني خائفه من امور انت اعلم بها فاجعل الطمانينه تسري بجسدي واتم اموري بتوفيقك لي وكن معي ياحي ياقيوم.	hope
8840	ألا تقرؤو الطاريخ ..بأن ذا القرنين ماغربي و قد فتح مكّة ..و بأن المغاربة هم الذين هزمو فرعون و أغرقوه في البحر سيرو اقراو الطاريخ ثم تقولون المغريب يحقيد علينا، من أطلق الثورة الجزائرية أليست المغريب ،حيث دخل الحسن الثاني إلى جبل الأوراس و أطلق أول رصاصة. [emojies] [URL] [USER]	hate
5472	و صحيح دي هتبقي اول مره في تاريخ الاولمبياد ان الافتتاح يبقي علي ملعب مش هيبقي عليه منافسات العاب القوي ، الافتتاح علي الماراكانا	not_applicable
3587	أُحِبُّ أَنْ أَضْحَكَ وَأَسْتَمْتِعَ بِالنُّكْتَةِ الْمُضْحِكَةِ.	hope
4127	اللهم أنزل دفء رحمتك على من لا مأوى لهم من برد الشتاء اللهم ابسط رحمتك عليهم وارزقهم الامن والامان	hope
3341	ضابطان في الجيش الأمريكي يتبادلان ممارسة السكس على بناتهم المراهقات المزز و أحلى عقاب بالنيك [URL]	not_applicable
135	من باب التنكيد على المزورين.. #الغبانة_هندية #المزمار_افريقي وماتمثلنا كسعوديين[emojies] [USER] RT [URL]	not_applicable
8092	اسفين سيادة الرئيس لقد ظلمناك يوم ساعدنا على تغييرك هاهم بحافل الشمال وبقايا عفاش وال الأحمر والحوثيين يتحدون ضد شعبنا الجنوبي بسبب اننا فتحنا ابوابنا لشله من الفاسدين الحاقدين يتحكمون بمصيرنا [URL]	not_applicable
5574	واصلي يامجموعه نايف بن خالد عل بركه الله مدعومه بالدعاء والتوفيق من متابعيك	hope
5546	احنا ملتزمين الصمت اهو وقاعدين نتفرج علي الاولمبياد لحد ما الماتش يخلص [emojies]	not_applicable
9643	سَقَمي في الحُبِّ عَافِيَتِي وَوُجُودِي فِي الهَوَى عَدَمِي وَعَذَابٌ تَرتَضُونَ بِهِ فِي فَمِي أحلَى مِنَ النِّعَمِ مَا لِضُرِّ فِي مَحَبَّتِكُم عِندَنا والله مِن أَلَمِ	not_applicable
6895	لما سبيت بنات كسرى يزدجرد احترمهن عمر رضي الله عنه وأشار عليهن أن تختار كل واحدة منهن واحد من أبناء الصحابة فاختارت احدهن الحسين بن علي واختارت الثانية محمد بن أبي بكر واختارت الثالثة عبدالله بن عمر، وهذا الأمر واقع على الرجال والنساء، ومازالت الدول	not_applicable

Table VI: Summary of the supplementary manual agreement check on 20 Task 1 samples.

Panel A: Overall agreement		
Metric	Value	Description
Samples	20	Random Task 1
Annotators	3	Independent
Krippendorff's α	0.326	Nominal
Panel B: Pairwise agreement		
Annotator pair	κ	Agreement
M1 vs M2	0.260	0.526
M1 vs M3	0.356	0.579
M2 vs M3	0.305	0.550

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[4] M. Tonneau, D. Liu, S. Fraiberger, R. Schroeder, S. Hale, and P. Röttger, "From languages to geographies: Towards evaluating cultural bias in hate speech datasets," in *Proceedings of the 8th Workshop on Online Abuse and Harms (WOAH 2024)*, Y.-L. Chung, Z. Talat, D. Nozza, F. M. Plaza-del Arco, P. Röttger, A. Mostafazadeh Davani, and A. Calabrese, Eds. Mexico City, Mexico: Association for Computational Linguistics, Jun. 2024, pp. 283–311. [Online]. Available: https://aclanthology.org/2024.woah-1.23

[5] L. Derczynski, K. Bontcheva, and I. Roberts, "Broad twitter corpus: A diverse named entity recognition resource," in *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, 2016, pp. 1169–1179.

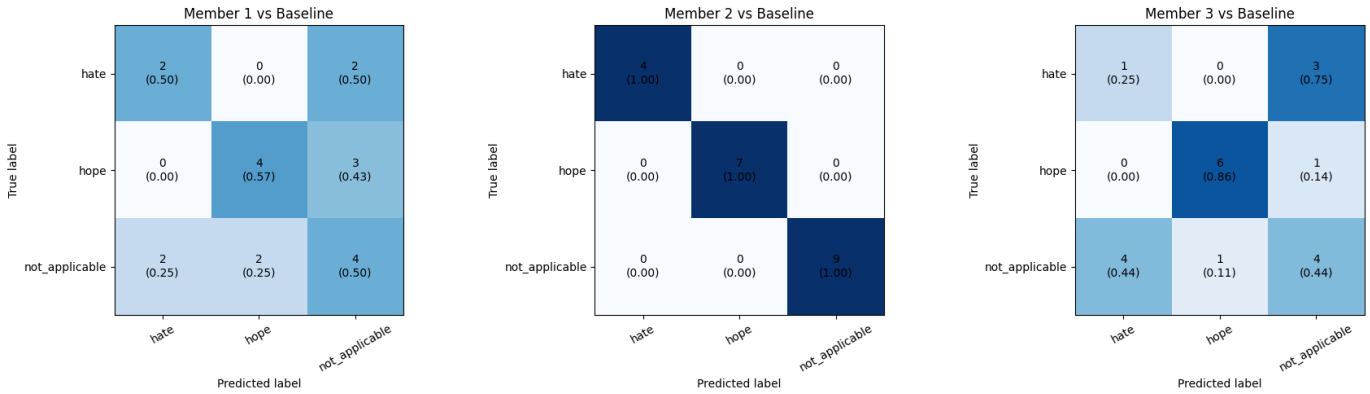

Figure 5: Confusion matrices comparing each annotator's labels with the MAHED Task 1 gold labels. The figure shows that most disagreements occur between *hate* and *not_applicable*, and between *hope* and *not_applicable*.